

High-speed Mach-Zehnder modulator by sputtered thin-film lithium niobate

Hiroki Hara, Shigeru Mieda, Katsumi Kawasaki, Takashi Kikukawa, Hiroshi Take, Tetsuya Mino, Atsushi Shimura, Yasuhiro Takagi, Hideaki Fukuzawa

Advanced Products Development Center, Technology & Intellectual Property HQ, TDK Corporation, Japan Hiroki.01.Hara@tdk.com

Abstract We successfully demonstrate high-speed data transfer at 192 Gbaud under four-level pulse amplitude modulation (PAM4), achieving a net bit rate of 384 Gbps using a sputtered thin-film lithium niobate (TFLN) Mach-Zehnder modulator. The sapphire substrate plays a key role in enabling this high performance without relying on bulk lithium niobate: ©2026 The Author(s)

Introduction

High-speed data transfer is a key requirement for next-generation optical interconnects, since higher data rate per-lane is necessary to reduce system complexity. In response to the rapid increase in data center traffic, the development of ultra-high-speed and mass-producible optical modulators has become an urgent priority. In order to achieve such high-speed data transfer, the realization of 400 Gbit/sec (Gbps) per lane by scalable solution is critically important without increasing fabrication cost. Accordingly, expanding the capacity of optical transfer systems now requires a significant increase in modulator bandwidth.

To address these requirements, silicon photonics is currently recognized as a leading platform due to its compatibility with existing complementary metal-oxide-semiconductor (CMOS) processes, and silicon (Si) modulators^[1-3] have been extensively researched for high-density integration. While Si modulators promise excellent scalability, they may face challenges in achieving the bandwidth required for future high-baud-rate systems. Additionally, indium phosphide (InP) modulators^[4-6] provide excellent high-speed performance and are already in practical use. However, both high linearity and thermal management for broadband operation remain key challenges for InP modulators.

In this context, lithium niobate (LiNbO_3 , LN), which possesses superior electro-optic properties, has recently regained significant attention^[7-18]. In particular, thin-film lithium niobate (TFLN) modulators have demonstrated high-speed communications exceeding 400 Gbps^[11]. However, conventional TFLN wafers were prepared by bonding methods using bulk-LN, which involves complex manufacturing processes with high-cost and it is very difficult to scale up to large-diameter to satisfy rapid growing demand for AI data center^[7-14]. Therefore, there is a strong demand for scalable TFLN fabrication methods.

One such potential alternative is sputtering method by physical vapor deposition equipment.

Notably, sputtering allows for deposition on large-diameter wafers and offers the potential for significant cost reductions, and this process has been widely used for semiconductor devices with huge scalability. In order to realize excellent scalability similar to semiconductor devices, we have developed TFLN by sputtered method^[15-18].

In this study, by using sputtered TFLN Mach-Zehnder modulator (MZM), high-speed data transfer of 192 Gbaud four-level pulse amplitude modulation (PAM4) with a net bit rate of 384 Gbps were demonstrated. This result shows that sputtered TFLN can be a scalable solution for next-generation data center optical interconnects.

Design and fabrication of devices

Instead of the conventional bonding method using bulk-LN, the sputter deposition method on sapphire substrate was employed^[15-18]. The X-ray diffraction analysis identified the sputtered film as LiNbO_3 based on the peak at $2\theta=38.9^\circ$, corresponding to the (006) reflection. The corresponding rocking curve exhibited a full width at half maximum of 0.05° . These results indicate that the sputtered TFLN exhibits high crystallinity.

Figure 1(a) shows a schematic top view of the sputtered TFLN MZM. Direct current (DC) bias electrodes run along the two LN waveguides, while radio frequency (RF) signal electrodes are placed in the electro-optic (EO) interaction region.

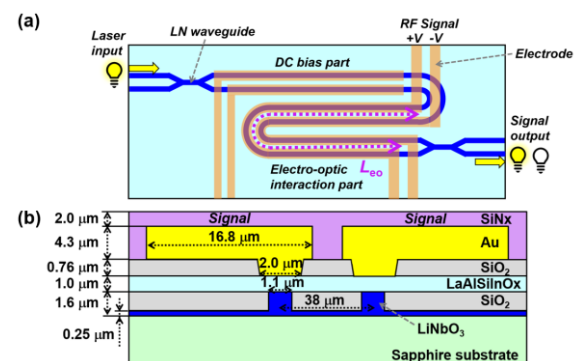


Fig. 1: Schematic figure of the TFLN MZM device with differential-driving electrode. (a) Top view image. (b) Cross-section view image of the EO interaction part.

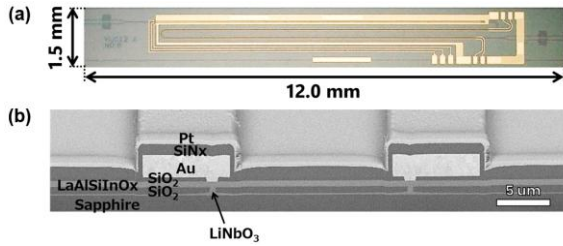


Fig. 2: (a) Die and (b) cross-sectional SEM image of the EO interaction region of the sputtered TFLN MZM.

This region, where the LN waveguides overlap the RF signal electrodes, has a length L_{eo} . The RF signal is differentially applied to the two signal electrodes, with its positive and negative components propagating along the respective electrodes. Figure 1(b) shows a schematic cross section of the EO interaction region, and Fig. 2 shows (a) the fabricated sputtered TFLN MZM and (b) its cross-sectional scanning electron microscopy (SEM) image.

EO measurement result

The modulation characteristics of the sputtered TFLN MZMs were measured using transverse magnetic (TM) polarized light at a wavelength of 1,310 nm in the O band. At a frequency of 100 Hz, the measured half-wave voltages (V_{π}) were 2.6, 3.2, and 3.8 V for L_{eo} of 16, 14, and 12 mm, respectively. Furthermore, the static extinction ratio (ER) was approximately 20 dB, indicating a high on-off contrast and demonstrating the excellent EO modulation capability of LN.

Figure 3 shows the measured EO frequency responses of the sputtered TFLN MZMs, normalized to the response at 1 GHz. A setup comprising an opto-electronic network analyser broadband VNA system ME7838D4 70 kHz to 145GHz, O/E calibration module MN4765B-0112 70 kHz to 110 GHz with 1310 and 1550 nm wavelength coverage, and 145 GHz millimeter-wave module MA25300A (Anritsu corporation) was used to measure the EO bandwidth up to 110 GHz. The 3-dB EO bandwidths were 70, 80, and 90 GHz for L_{eo} of 16, 14, and 12 mm, respectively. The corresponding 6-dB EO bandwidths were 85, 95, and 105 GHz, respectively. These results indicate that the EO bandwidth increased as L_{eo} decreased. This improvement is attributed to the

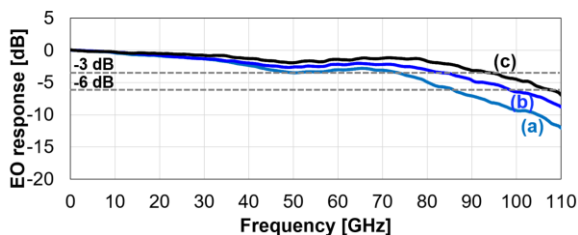


Fig. 3: Measured EO responses of the sputtered TFLN MZMs for L_{eo} of (a) 16 mm, (b) 14 mm, and (c) 12 mm, normalized to the response at 1 GHz.

lower microwave loss associated with the shorter electrodes.

High-speed data transfer

High-speed data transfer tests were performed using the sputtered TFLN MZM with $L_{eo} = 16$ mm, which exhibited the lowest V_{π} . A DCA-X sampling oscilloscope equipped with an N1032A module (Keysight Technologies) was used to measure the PAM4 eye diagrams, as shown in Fig. 4. Measurements were performed up to 384 Gbps using a short stress pattern random quaternary (SSPRQ) pattern under differential drive with a 1.4 V_{pp} voltage swing, and the transmitter and dispersion eye closure for quaternary (TDECQ) was evaluated from the measured waveforms.

Figure 5 shows the measured PAM4 eye diagrams and corresponding ER and TDECQ results. At 112 Gbaud (224 Gbps), an ER of 3.9 dB was obtained even without transmitter (Tx) and receiver (Rx) equalization. The TDECQs were 1.2 and 0.8 dB using 15-tap feed-forward equalizer (FFE) + 1-tap decision-feedback equalizer (DFE) and 51-tap FFE + 21-tap DFE, respectively. At 192 Gbaud (384 Gbps), an ER of 3.8 dB was obtained using a 5-tap Tx FFE without Rx equalization, while the corresponding TDECQs were 2.4 and 1.2 dB using 15-tap FFE + 1-tap DFE and 51-tap FFE + 21-tap DFE, respectively. These results show that 384 Gbps PAM4 data transfer was successfully demonstrated by using the sputtered TFLN MZM. Note that clear eye openings were obtained under various equalization conditions, even with relatively few equalizer taps.

Discussion

High-speed TFLN modulators have been widely studied, but none of these studies have demonstrated a modulator based on sputtered TFLN. Therefore, we compare the performance with bulk-LN bonding TFLN modulator. For example, 450 Gbps^[11], 224 Gbps^[13], and 144 Gbps^[14] have been reported by conventional TFLN modulators using bulk-LN bonding. By comparing with these results, our result of 384 Gbps clearly indicates that the sputtered TFLN MZMs offer ultra-high-speed communication capabilities comparable favorably to those of bulk-bonded films. Crucially, this high-speed transfer was enabled by the wide EO bandwidth of the sputtered TFLN MZMs

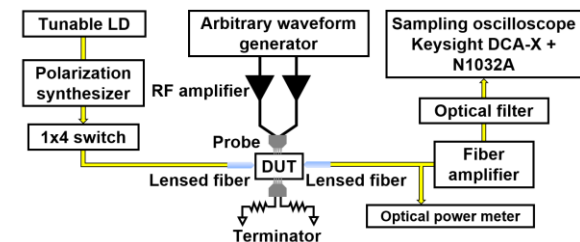


Fig. 4: Eye diagrams measurement setup.

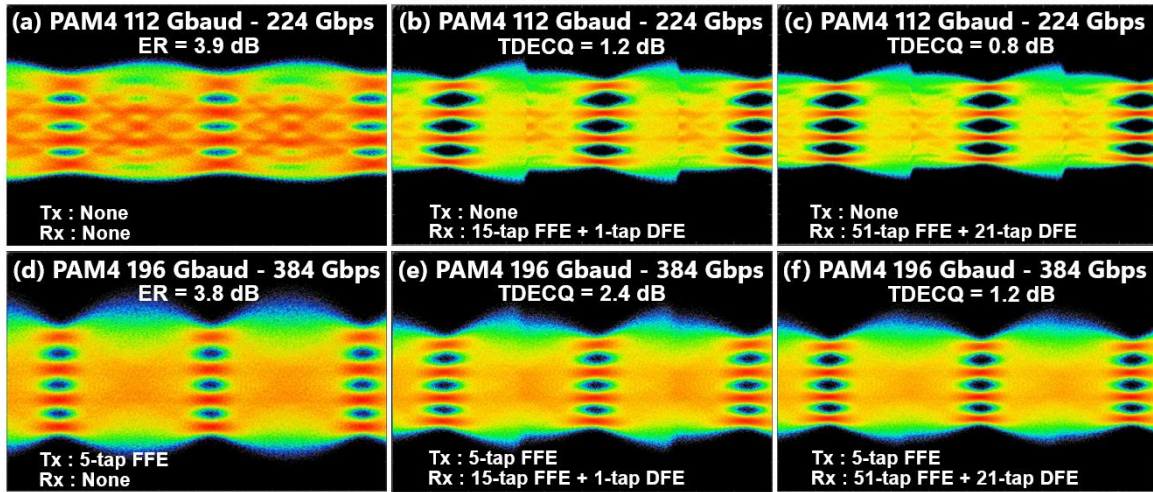


Fig. 5: Measured PAM4 eye diagrams of the sputtered TFLN MZM ($L_{\text{eo}} = 16$ mm): (a) 112 Gbaud (224 Gbps) without equalization; (b) and (c) the same data rate with 15-tap Rx FFE + 1-tap Rx DFE and 51-tap Rx FFE + 21-tap Rx DFE, respectively; (d) 192 Gbaud (384 Gbps) with 5-tap Tx FFE and without Rx equalization; and (e) and (f) the same data rate with 5-tap Tx FFE and 15-tap Rx FFE + 1-tap Rx DFE and 51-tap Rx FFE + 21-tap Rx DFE, respectively.

without using bulk-LN. This result is surprising because our TFLN MZMs were fabricated by sputtered TFLN. The reasons of successful demonstration for such high-speed data transfer are discussed below.

The clear PAM4 eye openings shown in Fig. 5 can be attributed to the following two factors: First, the sapphire substrate possesses superior dielectric properties in terms of both loss tangent and permittivity. Specifically, the extremely small loss tangent of the sapphire substrate ($\tan\delta < 0.0001^{[19]}$) plays a critical role in reducing RF signal loss. In contrast, conventional bulk-bonded TFLN on silicon-on-insulator (SOI) substrates suffers from the larger loss tangent of the Si substrate ($\tan\delta = 0.002^{[20]}$), which is 20 times higher than that of sapphire. Furthermore, the low relative permittivity of the sapphire substrate ($\epsilon_r = 9.39^{[19]}$) is also advantageous for EO bandwidth. A higher substrate permittivity increases the effective microwave refractive index, which in turn decreases the microwave phase velocity. Because the permittivity of Si is large ($\epsilon_r = 11.68^{[20]}$), the microwave velocity becomes excessively low; consequently, the optical signal velocity must be artificially reduced to achieve the velocity matching required for a wide EO bandwidth. To overcome both the large loss tangent and high permittivity of the Si substrate, TFLN on SOI requires complex device structures, such as etching air holes in the Si substrate beneath the LN waveguides^[10,12]. While fabricating such undercut structures significantly increases the manufacturing complexity and cost, our sputtered TFLN on sapphire completely avoids this complex process and remains scalable up to 12-inch wafers.

Second, the crystalline quality of TFLN by sputtering was sufficiently good enough to realize high-speed data transfer. While poor crystallinity

in TFLN leads to increased dielectric loss and optical propagation loss, our high-crystallinity sputtered TFLN suppresses these losses.

The PAM4 eye diagrams shown in Fig. 5 (d), (e) and (f) were measured under differential drive with a $1.4 V_{pp}$ voltage swing. Although the applied $1.4 V_{pp}$ drive signal was much lower than the measured V_{π} of 5.2 V, the clear PAM4 eye openings were observed due to the high ER of TFLN MZM. As a result, clearly distinguishable PAM4 eye levels were obtained even by using the V_{pp} much lower than V_{π} . This is a great benefit for using TFLN MZM compared to Si modulators.

Receiver-side integration is also essential for practical optical interconnects. Combining this ultrafast transmitter with high-speed photodetectors^[21-22] could enable fully integrated, next-generation optical communication systems.

Conclusions

In conclusion, high-speed data transfer was successfully demonstrated using a TFLN MZM fabricated by sputtering. A net bit rate of 384 Gbps was achieved by using PAM4 signalling at a symbol rate of 192 Gbaud. This successful high-speed data transfer compares favorably to that of conventional bulk-bonded TFLN modulators, while the sputtered TFLN can offer much better scalability owing to its simplified fabrication process. These results support sputtered TFLN as a scalable solution for next-generation optical interconnects and a promising platform for broadband electro-optic modulators.

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